

Bayesian inversion using adaptive Polynomial Chaos Kriging within Subset Simulation

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Outline

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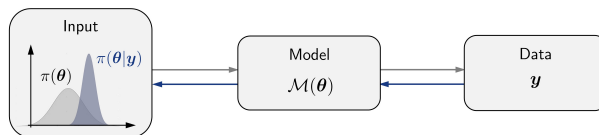
Bayesian inference for inverse problems

- Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space
- Let $\mathcal{M} : \mathcal{D}_\theta \subseteq \mathbb{R}^d \longrightarrow \mathbb{R}^n$ a deterministic **model** with random **input parameters** $\Theta : \Omega \longrightarrow \mathcal{D}_\theta$
- The level of knowledge about Θ is summarized by a **prior** density $\pi(\theta)$
- Given observed data $\mathbf{y} \in \mathbb{R}^n$, the **posterior** density of input parameters may be derived with Bayes' theorem:

$$\pi(\theta|\mathbf{y}) = \frac{\pi(\theta)\mathcal{L}(\theta;\mathbf{y})}{\mathcal{Z}}$$

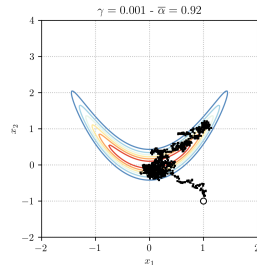
where $\mathcal{L}(\theta;\mathbf{y})$ is the **likelihood function**, and $\mathcal{Z}$ the **model evidence**:

$$\mathcal{Z} = \int_{\mathcal{D}_\theta} \mathcal{L}(\theta;\mathbf{y})\pi(\theta)d\theta$$



Bayesian computations

- Bayesian computations are often based on MCMC sampling techniques
(Robert & Casella, 2004; Gamerman & Lopes, 2006)
- MCMC typically requires a large number of likelihood evaluations
- The forward model $\mathcal{M}$ may be costly
- Need to accelerate Bayesian computations



Example of a MCMC run

Surrogate modeling for accelerated Bayesian computations

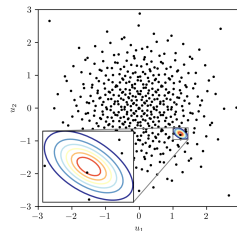
- In order to accelerate Bayesian computations, the forward model $\mathcal{M}$ is usually replaced by a **surrogate model** $\hat{\mathcal{M}}$:

$$\mathcal{M}(\boldsymbol{\theta}) \approx \hat{\mathcal{M}}(\boldsymbol{\theta})$$

- In this context, **prior-based** surrogate models are widely used, in order to device a global approximation of the forward model on the prior support

(Marzouk et al., 2007; Sraij et al., 2015)

- However, such approaches may fail if data are strongly informative...
(Lu et al., 2015; Yan & Zhou, 2019)
- ... and/or if the forward model is strongly nonlinear on the prior support
- Local accuracy near informative zones may not be guaranteed, even with a large experimental design size



Prior-based experimental design vs. concentrated posterior

Motivations and proposed methodology

- Main goal: construct a surrogate model with an **adaptive** experimental design, by selecting enrichment points near informative zones
- Proposition of an approach based on the so-called **BUS** framework (*Bayesian Updating with Structural reliability methods*), which reformulates Bayesian inference into a Structural Reliability (SR) problem
(Straub & Papaioannou, 2015)
- BUS is used as a basis for devising **active-learning** schemes inspired from SR methods
(Bichon et al., 2008; Echard et al., 2011; Moustapha et al., 2022)
- Use of **Polynomial Chaos Kriging (PCK)** surrogates: (Schöbi et al., 2015, 2017)
 - ▶ Provide a global approximation on the prior support with Polynomial Chaos Expansions (PCE)...
 - ▶ ...as well as local features provided by the Kriging part, for improving local accuracy near informative zones

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The BUS formulation

- Let $c > 0$ be a constant such that $c\mathcal{L}(\boldsymbol{\theta}; \mathbf{y}) \leq 1$, for all $\boldsymbol{\theta} \in \mathcal{D}_{\boldsymbol{\theta}}$
- Consider a random variable $P \sim \mathcal{U}([0, 1])$ independent from the parameters $\boldsymbol{\Theta}$
- One defines the following **failure event** in the augmented space $\mathcal{D}_{\boldsymbol{\theta}} \times [0, 1]$:

$$\mathcal{F} = \{(\boldsymbol{\theta}, p) \in \mathcal{D}_{\boldsymbol{\theta}} \times [0, 1] \mid p \leq c\mathcal{L}(\boldsymbol{\theta}; \mathbf{y})\}$$

- The posterior density $\pi(\boldsymbol{\theta}|\mathbf{y})$ then rewrites:

(Straub & Papaioannou, 2015)

$$\pi(\boldsymbol{\theta}|\mathbf{y}) = \frac{1}{\mathbb{P}_{\mathcal{F}}} \int_0^1 \mathbf{1}_{\mathcal{F}}(\boldsymbol{\theta}, p) dp$$

where $\mathbb{P}_{\mathcal{F}}$ is the **failure probability** defined by:

$$\mathbb{P}_{\mathcal{F}} = \mathbb{P}((\boldsymbol{\Theta}, P) \in \mathcal{F}) = \int_{\mathcal{D}_{\boldsymbol{\theta}}} \int_0^1 \mathbf{1}_{\mathcal{F}}(\boldsymbol{\theta}, p) \pi(\boldsymbol{\theta}) dp d\boldsymbol{\theta} = cZ$$

The BUS formulation

- Sampling from the posterior $\pi(\boldsymbol{\theta}|\mathbf{y})$ is tantamount to solving the reliability problem associated to the following **Limit State Function (LSF)**:

$$\mathcal{G}(\boldsymbol{\theta}, p) = p - c\mathcal{L}(\boldsymbol{\theta}; \mathbf{y})$$

$$\mathcal{F} = \{(\boldsymbol{\theta}, p) \in \mathcal{D}_{\boldsymbol{\theta}} \times [0, 1] \mid \mathcal{G}(\boldsymbol{\theta}, p) \leq 0\}$$

- Samples that lie in $\mathcal{F}$ are following the posterior distribution

(Straub & Papaioannou, 2015)

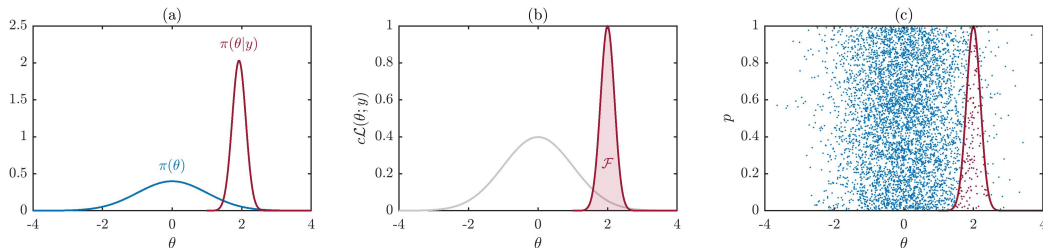


Illustration of the BUS framework: (a) Bayesian inference; (b) Definition of the failure event $\mathcal{F}$; (c) Solving the equivalent reliability problem through MCS

Subset Simulation (SuS) within BUS

- Use of the well-known Subset Simulation (SuS) technique for solving the equivalent SR problem:

$$\mathbb{P}_{\mathcal{F}} = \mathbb{P} \left(\bigcap_{i=0}^r \mathcal{F}_i \right) = \prod_{i=1}^r \mathbb{P}(\mathcal{F}_i | \mathcal{F}_{i-1})$$

where $\mathcal{F} = \mathcal{F}_r \subset \mathcal{F}_{r-1} \subset \dots \subset \mathcal{F}_1 \subset \mathcal{F}_0$, with $\mathbb{P}(\mathcal{F}_0) = 1$

(Au & Beck, 2001)

- Definition of the subsets $(\mathcal{F}_i)_{0 \leq i \leq r}$:

$$\mathcal{F}_i = \{(\boldsymbol{\theta}, p) \in \mathcal{D}_{\boldsymbol{\theta}} \times [0, 1] \mid \mathcal{H}(\boldsymbol{\theta}, p) \leq h_i\}$$

where $0 = h_r < h_{r-1} < \dots < h_1 < h_0 = +\infty$ are threshold values, and $\mathcal{H}$ is a modified version of the LSF $\mathcal{G}$:

$$\mathcal{H}(\boldsymbol{\theta}, p) = \log(p) - \log(c) - \log(\mathcal{L}(\boldsymbol{\theta}; \mathbf{y}))$$

(DiazDelaO et al., 2017; Betz et al., 2018)

- The $(h_i)_{1 \leq i \leq r-1}$ are chosen such that the conditional probabilities $(\mathbb{P}(\mathcal{F}_i | \mathcal{F}_{i-1}))_{1 \leq i \leq r-1}$ are equal to a prescribed value $\mathbb{P}_t$ in average (with usually $\mathbb{P}_t = 10\%$)

Subset Simulation (SuS) within BUS

- Main steps of SuS procedure:
 - ▶ estimation of $\mathbb{P}(\mathcal{F}_1)$ through prior-based MCS
 - ▶ estimation of the conditional probabilities $(\mathbb{P}(\mathcal{F}_i|\mathcal{F}_{i-1}))_{2 \leq i \leq r}$ through MCMC sampling
- Use of a MCMC algorithm tailored for SuS

(Papaioannou et al., 2015)

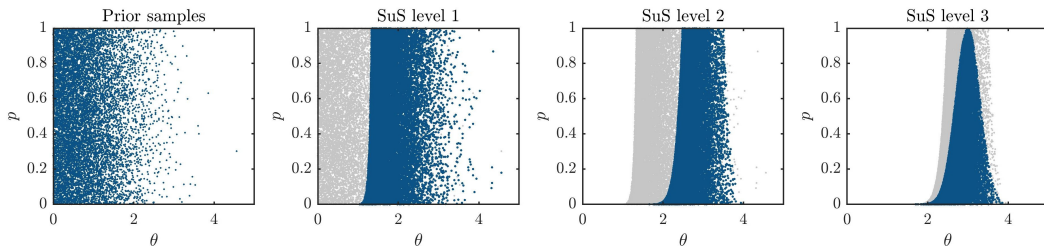


Illustration of SuS procedure, in the case $\pi(\theta) = \mathcal{N}(\theta|0, 1)$ and $\mathcal{L}(\theta; y) = \mathcal{N}(\theta|2, 0.2)$

The choice of the BUS normalizing constant

- The choice of the normalizing constant c plays a key role in the BUS-SuS procedure

- Optimal choice for c :

$$c_{\text{opt}}^{-1} = \max_{\theta \in \mathcal{D}_{\theta}} \mathcal{L}(\theta; \mathbf{y})$$

however, such value is usually not known in advance, and MLE may be intractable

- If $c^{-1} < c_{\text{opt}}^{-1}$, BUS-SuS samples may not follow the posterior distribution...
- ...otherwise, if $c^{-1} > c_{\text{opt}}^{-1}$, the overall performance of BUS-SuS may be altered, since $\mathbb{P}_{\mathcal{F}} \propto c$
- Solution adopted: on the fly adaptation of c throughout SuS procedure, by setting c as the maximum likelihood value on samples of the current SuS level

(DiazDelaO et al., 2017; Betz et al., 2018)

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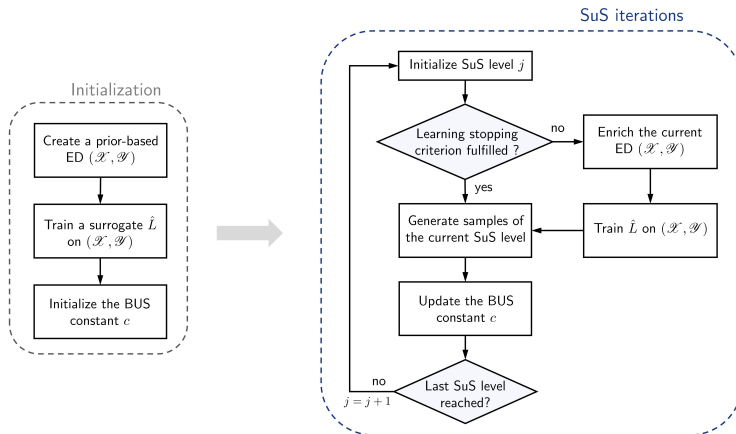
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Adaptive surrogate modeling procedure

- General framework of the proposed approach:

(D.R., Baroth, Briffaut, & Dufour, 2021)



PCK-based surrogate modeling

- Use of Polynomial Chaos Kriging (PCK) surrogate models:

(Schöbi et al., 2015, 2017)

$$L(\boldsymbol{\theta}) \approx \hat{L}(\boldsymbol{\theta}) = \sum_{\alpha \in \mathcal{A}} a_{\alpha} \psi_{\alpha}(\boldsymbol{\theta}) + Z(\boldsymbol{\theta})$$

where $(\psi_{\alpha})_{\alpha \in \mathcal{A}}$ is a truncated basis of multivariate polynomials, orthonormal w.r.t. the prior, and Z a zero-mean stationary Gaussian process

- ▶ Polynomial basis $(\psi_{\alpha})_{\alpha \in \mathcal{A}}$ determined through the LARS-based procedure of (Blatman & Sudret, 2011)
- ▶ $(\psi_{\alpha})_{\alpha \in \mathcal{A}}$ may then be entirely embedded in PCK as a Kriging trend (Sequential PCK), or iteratively in order to minimize a cross-validation error (Optimal PCK)

(Schöbi et al., 2015)

- ▶ PCK surrogates are built with the UQLab software

(Marelli & Sudret, 2014)

Learning stopping criterion

- Let $\mu_{\hat{L}}(\boldsymbol{\theta})$ and $\sigma_{\hat{L}}^2(\boldsymbol{\theta})$ be the prediction mean and variance of the PCK surrogate $\hat{L}$ at a point $\boldsymbol{\theta} \in \mathcal{D}_{\boldsymbol{\theta}}$
- Replacing the log-likelihood by its surrogate $\hat{L}(\boldsymbol{\theta})$ yields a surrogate LSF:

$$\hat{\mathcal{H}}(\boldsymbol{\theta}, p) = \log(p) - \log(c) - \hat{L}(\boldsymbol{\theta})$$

- $\hat{\mathcal{H}}$ is also a Gaussian process, with $\hat{\mathcal{H}}(\boldsymbol{\theta}, p) \sim \mathcal{N}(\log(p) - \log(c) - \mu_{\hat{L}}(\boldsymbol{\theta}), \sigma_{\hat{L}}^2(\boldsymbol{\theta}))$

- Use of the well-known **U learning function** based on surrogate LSF predictions:

(Echard et al., 2011)

$$U(\boldsymbol{\theta}, p) = \frac{|\mu_{\hat{\mathcal{H}}}(\boldsymbol{\theta}, p)|}{\sigma_{\hat{\mathcal{H}}}(\boldsymbol{\theta}, p)}$$

- Use of a standard stopping criterion: $\min U > \epsilon_U$ (with usually $\epsilon_U = 2$)

(Echard et al., 2011)

Point enrichment procedure

At each SuS level, before the beginning of MCMC sampling:

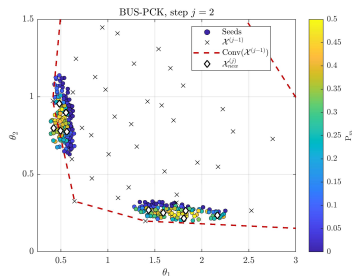
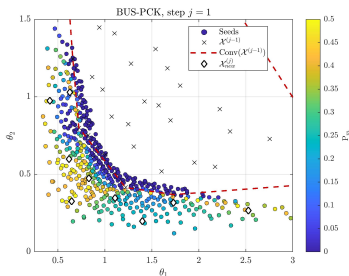
- The Markov chain seeds are ranked according to their **misclassification probability**:

(Bect et al., 2011)

$$\mathbb{P}_m(\boldsymbol{\theta}, p) = \Phi(-U(\boldsymbol{\theta}, p))$$

where Φ is the standard normal CDF

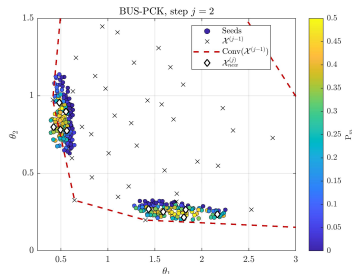
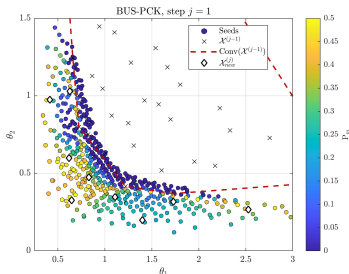
- If current ED is enriched by selecting the point which maximizes $\mathbb{P}_m$, or by using weighted k -means clustering in the case of multiple point enrichment



On the fly updating of the BUS scaling constant

At the end of each SuS level:

- The scaling constant c is updated by taking the maximum likelihood value on the current SuS samples
(DiazDelaO et al., 2017; Betz et al., 2018)
- Samples which are too far away from the current ED may corrupt the estimation of c
- Proposed solution: update c based on samples which lie in the **convex hull** of the current ED (Barber et al., 1996)



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Numerical tests cases: general settings

- Input parameters of the proposed approach: initial ED size K_0 , number of enrichment points per subset K_t
- Reference results produced with the BUS-SuS method
- Error measure on univariate posterior marginals: **averaged Jensen-Shannon-Divergence (aJSD)**

$$\overline{\Delta}_{\text{JS}}(\pi||\hat{\pi}) = \frac{1}{d} \sum_{i=1}^d \Delta_{\text{JS}}(\pi_i(\cdot|\mathbf{y}), \hat{\pi}_i(\cdot|\mathbf{y}))$$

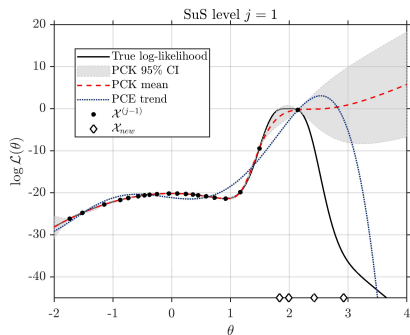
(Wagner et al., 2021)

where d is the input dimension, $\pi_i(\cdot|\mathbf{y}), \hat{\pi}_i(\cdot|\mathbf{y})$ are the reference and surrogate-based posterior marginals, and Δ_{JS} is the Jensen-Shannon Divergence (JSD)

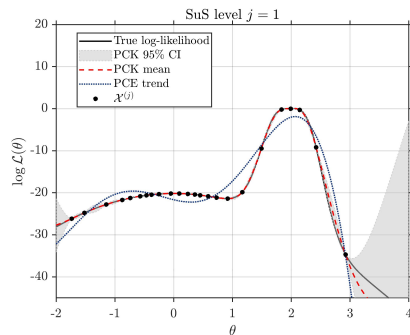
(Lin, 1991)

Example 1: 1D analytical function

- Example of a 1D analytical function: $\mathcal{M}(\theta) = 1 + \cos\left(\frac{\theta}{2}\right) + 3\exp(-4(\theta - 2)^2)$
- Model setup: $\Theta \sim \mathcal{N}(0, 1)$ and $\mathcal{L}(\theta; y) = \mathcal{N}(\mathcal{M}(\theta)|y, \sigma^2)$, with $y = \mathcal{M}(2)$ and $\sigma = 0.4$

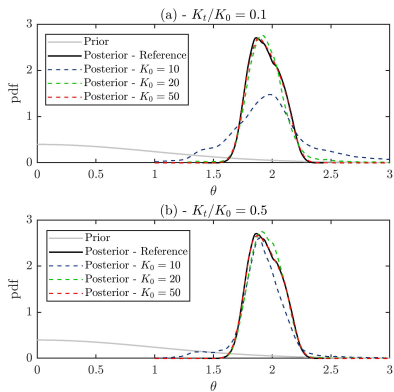


Log-likelihood at SuS level 1, before point enrichment

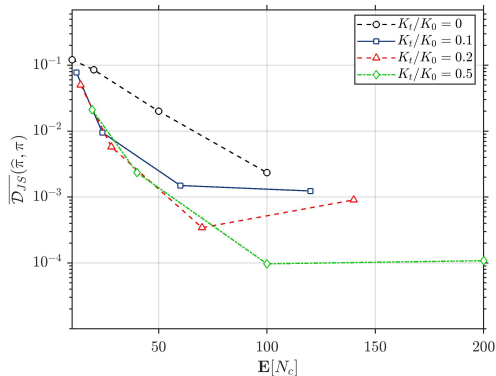


Log-likelihood at SuS level 1, after point enrichment

Example 1: 1D analytical function



Prior and posterior densities

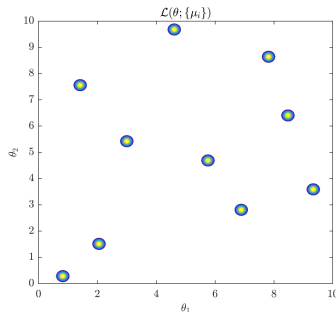


aJSD vs. average total number of model calls $\mathbb{E}[N_c]$

Example 2: 2D multi-modal posterior

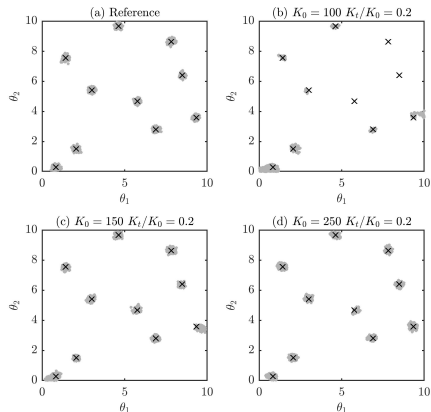
- Example of a multi-modal posterior distribution, with 10 distinct modes
- Prior: $\Theta \sim \mathcal{U}([0, 10]^2)$; Data: $\mathbf{y} = \{\mu_i\}_{1 \leq i \leq m}$, with $\mu_i \sim \mathcal{U}([0, 10]^2)$ and $m = 10$
- Likelihood: $\mathcal{L}(\theta; \mathbf{y}) = \frac{1}{m} \sum_{i=1}^m \mathcal{N}(\theta | \mu_i, \sigma^2)$

(Beck & Zuev, 2013)

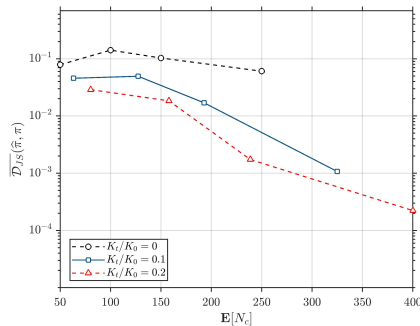


Contours of the likelihood $\mathcal{L}(\theta; \mathbf{y})$

Example 2: 2D multi-modal posterior



Posterior samples



aJSD vs. average total number of model calls $\mathbb{E}[N_c]$

Example 3: diffusion problem

- Example of a 1D diffusion problem with a random diffusivity field (Marzouk et al., 2007; Straub et al., 2016)
- Diffusion equation, with N sources with locations $(l_i)_{1 \leq i \leq N}$, strengths $(s_i)_{1 \leq i \leq N}$ and widths $(\sigma_i)_{1 \leq i \leq N}$:

$$\nabla (a(x) \nabla u(x)) + \sum_{i=1}^N \frac{s_i}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(l_i - x)^2}{2\sigma_i^2}\right) = 0$$

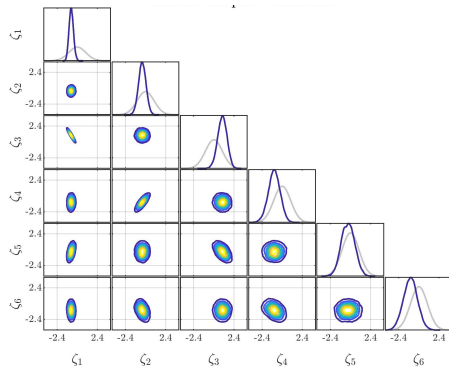
where $a(x)$ is the diffusivity field, and $u(x)$ the unknown field

- Prior log-diffusivity field: Gaussian stationary random field with exponential correlation kernel
- Discretization of $\log a(x)$ with a truncated Karhunen-Loève expansion, with $m = 10$ eigenmodes:

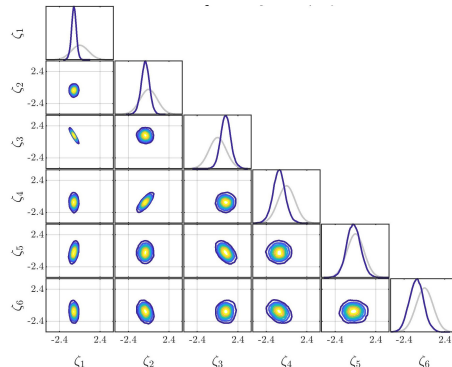
$$\log a(x) \approx \mu_{\log a} + \sum_{i=1}^m \sqrt{\lambda_i} \theta_i \phi_i(x)$$

- Inference of $\boldsymbol{\theta} = (\theta_i)_{1 \leq i \leq m}$ from measurements of the field $u(x)$ corresponding to a realization of $a(x)$

Example 3: diffusion problem

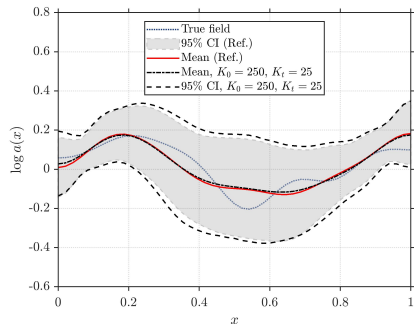


Reference posterior

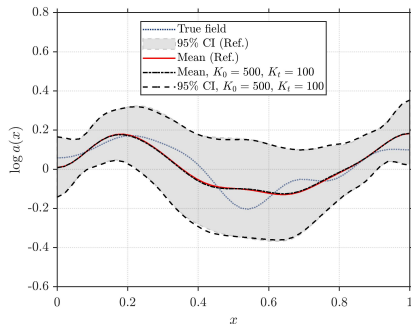


Surrogate-based posterior, $K_0 = 250$, $K_t = 25$

Example 3: diffusion problem

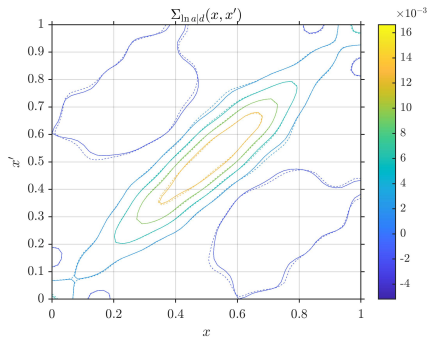


Posterior log-diffusivity field, $K_0 = 250, K_t = 25$

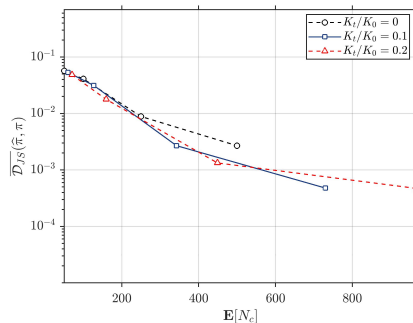


Posterior log-diffusivity field, $K_0 = 500, K_t = 50$

Example 3: diffusion problem



Posterior covariance kernel, $K_0 = 500$, $K_t = 100$ (Continuous lines: reference; dashed lines: surrogate-based)



aJSD vs. average total number of model calls $\mathbb{E}[N_c]$

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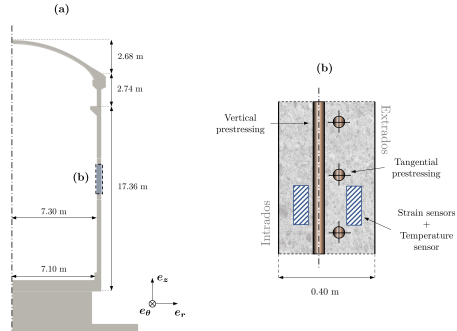
Conclusions

Case study: the VeRCoRs mock-up

- VeRCoRs mock-up: 1:3 scale double-walled Nuclear Containment Building (NCB) mock-up (Charpin et al., 2021)
- Well instrumented structure involving a large amount of monitoring data
- Structural (and local) leakage rate measured through periodic pressurization tests



Maquette VeRCoRs, EDF-LAB Les Renardières



(a) 2D view of VeRCoRs mock-up; (b) Focus on a standard zone

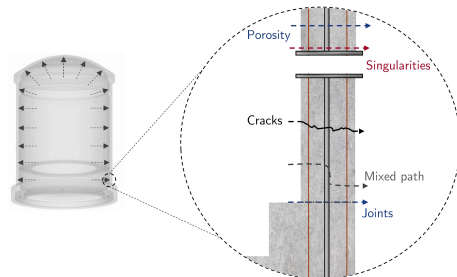
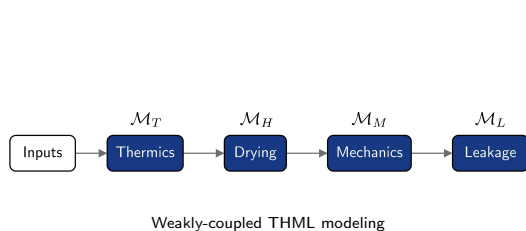
Modeling the physical behavior of NCB

- Modeling of the spatio-temporal evolution of thermal, hydrous, mechanical and transfert quantities of interest (e.g. water saturation, strains, leakage rate)

- Use of a **weakly-coupled** thermo-hydro-mechanical & leakage (THML) model:

(D.R., Bouhjiti, et al., 2021)

$$\mathcal{M} = \mathcal{M}_L \circ \mathcal{M}_M \circ \mathcal{M}_H \circ \mathcal{M}_T$$



Phenomenology of leaks through the inner wall during a pressurization test

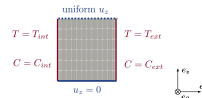
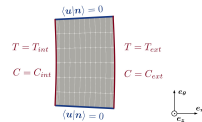
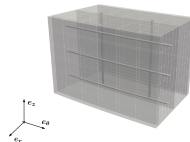
Finite element model and input parameters

- Use of **Representative Structural Volumes (RSV)** for modeling the behavior of the mock-up

(D.R., Baroth, Briffaut, Dufour, Monteil, et al., 2021)

Identification of 9 influent model parameters:

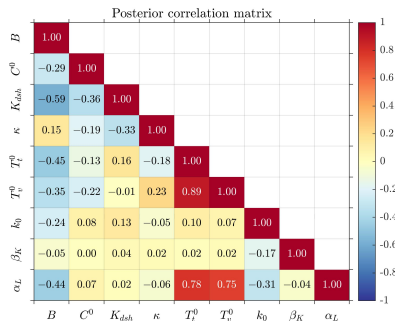
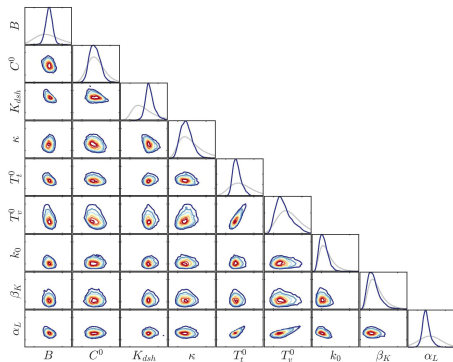
- ▶ 2 thermo-hydrous parameters: (B, C^0)
- ▶ 4 mechanical parameters:
 $(K_{dsh}, \kappa, T_t^0, T_v^0)$
- ▶ 3 leakage parameters: (k_0, β_K, α_L)



Finite element mesh of VeRCoRs standard zone RSV & boundary conditions

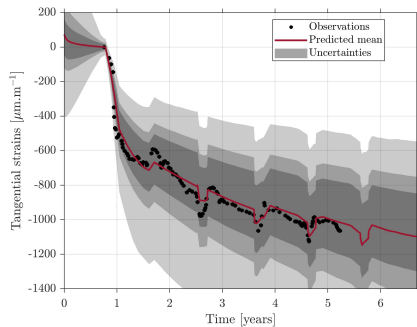
Bayesian updating of model parameters

- Bayesian updating of model parameters using strain and leakage monitoring data

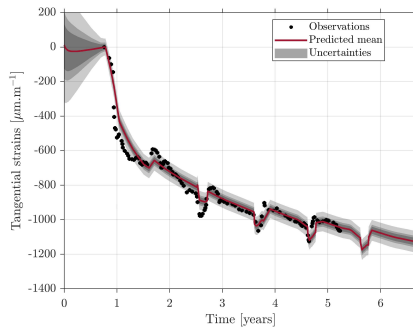


Probabilistic delayed strains predictions

■ Probabilistic predictions of tangential delayed strains of VeRCoRs standard zone



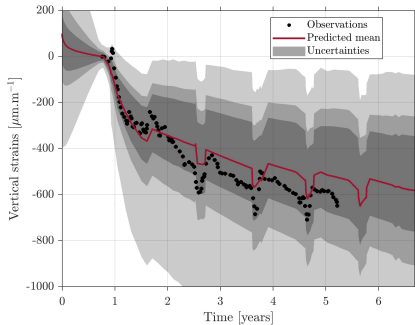
Prior tangential strains of VeRCoRs standard zone (Mean & 50, 70, 95% CI)



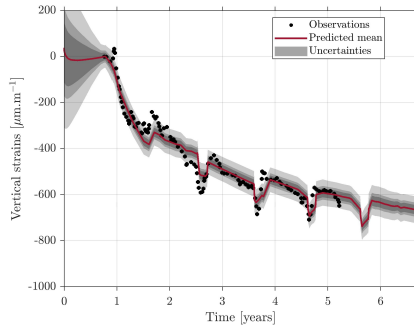
Posterior tangential strains of VeRCoRs standard zone (Mean & 50, 70, 95% CI)

Probabilistic delayed strains predictions

- Probabilistic predictions of vertical delayed strains of VeRCoRs standard zone



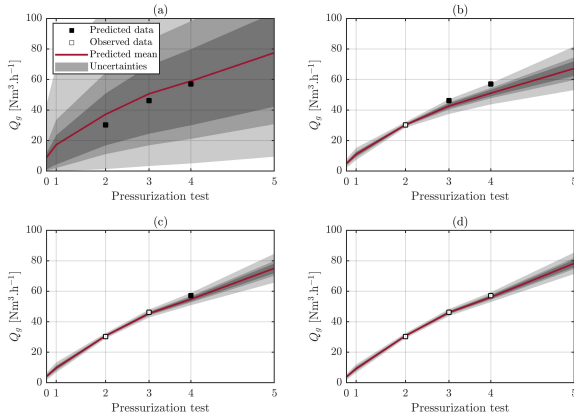
Prior vertical strains of VeRCoRs standard zone (Mean & 50, 70, 95% CI)



Posterior vertical strains of VeRCoRs standard zone (Mean & 50, 70, 95% CI)

Probabilistic global leakage predictions

- Probabilistic predictions of the **global leakage rate** of VeRCoRs mock-up



Predictions of the VeRCoRs global leakage rate
(Mean & 50, 70, 95% CI): (a) prior predictions;
(b) after Test 1; (c) after Test 2; (d) after Test 3

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- Development of a Bayesian updating approach combining the BUS framework with adaptive surrogate models
- The proposed method is particularly efficient for sampling posteriors with "localized" behaviors (e.g. concentrated posteriors, multi-modalities)
- Considerable reduction of the computational cost required by Bayesian sampling techniques ($\sim 10^2 - 10^3$ for surrogate-based calculations)
- The proposed approach may be extended in order to update probabilities of failure in a Bayesian framework (possible application to real NCB with regulatory leakage thresholds) (Straub et al., 2016)
- Improvement perspectives related to the choice of the learning stopping criterion, and to the handling of high dimensionalities (> 20)

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